Abstract

This paper presents a systematic investigation of unimodal and multimodal deep learning approaches for emotion classification, comprising eight model configurations. Unimodal text experiments show that fine-tuned DistilBERT achieves 91.8% accuracy on the Twitter Emotions dataset. Unimodal image experiments on OAHEGA show that ResNet-18 transfer learning (79.6%) substantially outperforms a from-scratch CNN (55.4%). A controlled multimodal experiment demonstrates that dataset alignment is the dominant factor in cross-modal learning: randomly paired unaligned data yields Cohen's Kappa of 0.00, while the synchronized MELD dataset enables genuine learning. Two MELD multimodal variants are evaluated — CNN encoder (42.2%, Kappa 0.29) and ResNet-18 encoder (47.6%, Kappa 0.31) — confirming that transfer learning benefits extend into the multimodal fusion setting.